

HTTP Fundamentals & Interception Analysis

Technical Overview of HTTP/1.1 Mechanics & Practical Interception on DVWA

1. Understanding HTTP Architecture

Hypertext Transfer Protocol (HTTP) is an application-layer protocol designed for transferring distributed, collaborative hypermedia presentation information. Operating primarily over TCP/IP (port 80 for unencrypted HTTP, port 443 for HTTPS), HTTP follows a stateless client-server model.

1.1 HTTP Request Components

A standard HTTP request sent by a client (e.g., a web browser or security tool) consists of three primary elements:

- **Request Line:** Contains the HTTP method (GET, POST, PUT, DELETE, etc.), the target Request-URI, and the protocol version (e.g., HTTP/1.1).
- **Request Headers:** Key-value pairs providing metadata about the client, session, content encoding, and host configuration (e.g., Host, User-Agent, Cookie, Content-Type).
- **Message Body:** Optional data payload submitted with requests such as POST or PUT, containing form parameters, JSON data, or file uploads.

1.2 Common HTTP Methods

Method	Idempotent	Description & Typical Use Case
GET	Yes	Requests a representation of the specified resource. Should only retrieve data and have no other effect.
POST	No	Submits data to be processed to a specified resource, often resulting in a change in state or side effects on the server.
PUT	Yes	Replaces all current representations of the target resource with the request payload.
DELETE	Yes	Deletes the specified target resource from the server.

2. Intercepting HTTP Traffic via Burp Suite on DVWA

Burp Suite operates as a Man-in-the-Middle (MitM) HTTP proxy positioned between the browser and the target server. By intercepting requests on Damn Vulnerable Web Application (DVWA), security analysts can inspect, modify, and analyze raw HTTP traffic prior to server processing.

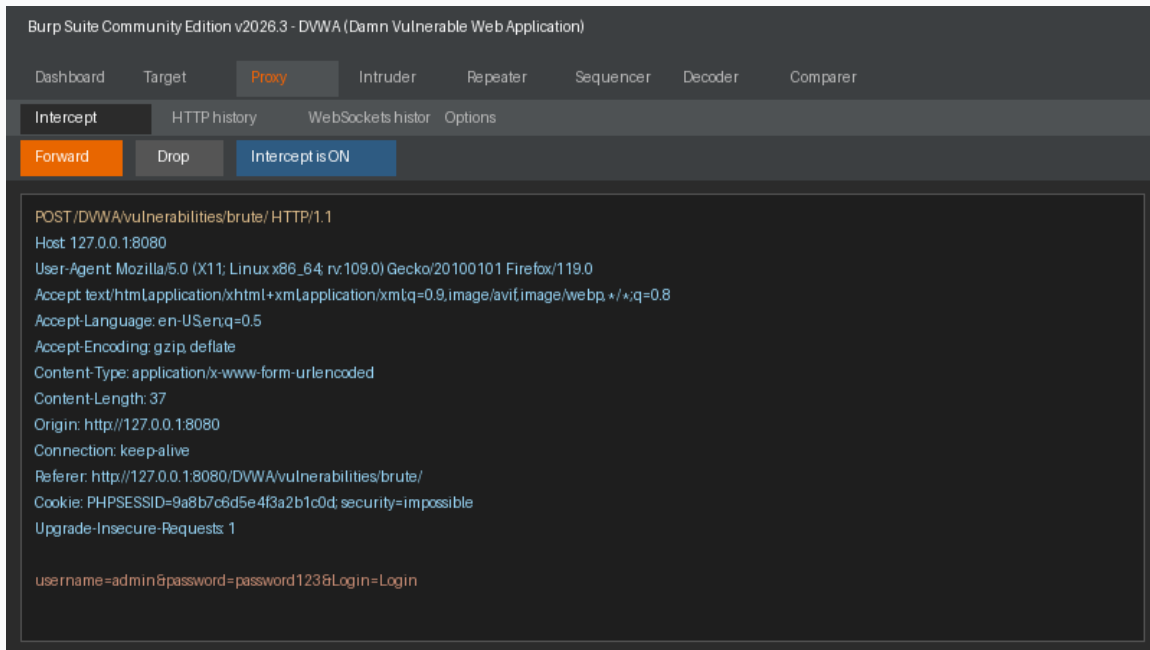


Figure 1: Burp Suite Proxy Intercept capturing a POST authentication request to DVWA Brute Force module.

2.1 Technical Breakdown of Captured Request

An analysis of the intercepted HTTP packet depicted in Figure 1 highlights several key security parameters:

- **Target Endpoint:** The request targets POST /DVWA/vulnerabilities/brute/, indicating form submission rather than resource retrieval.
- **Session State (Cookies):** The request transmits a PHP Session ID (PHPSESSID) and security level configuration (security=impossible). These key-value pairs maintain user session state across stateless HTTP calls.
- **Content Headers:** Content-Type: application/x-www-form-urlencoded specifies how payload parameters are formatted in the HTTP request body.
- **Payload Inspection:** The message body contains cleartext parameters (username=admin&password=password123&Login=Login), illustrating why transport layer encryption (HTTPS/TLS) is vital to prevent credential exposure.

Key Security Note: Intercepting HTTP requests allows security assessors to tamper with parameters, test for Broken Access Control, SQL Injection, and Cross-Site Request Forgery (CSRF) before the request reaches the server.

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Task-1: Walking An Application

In this room you will learn how to manually review a web application for security issues using only the in-built tools in your browser. More often than not, automated security tools and scripts will miss many potential vulnerabilities and useful information.

Share your win with your peers
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Burp Suite: Repeater completed!

neharehan2020 completed another room on their cyber security journey.

Completed tasks

9

Points earned

56

Streak

2

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